
CYBERSECURITY PRACTICAL TRAINING

PROJECT 1

Advanced Port Scanner

Network Reconnaissance & Vulnerability Assessment Tool

Cybersecurity Practical Training Program

Student: kalibaba | Kali Linux – VMware Environment

Date: July 7, 2026 | Deadline: July 15, 2026

Instructor: Cybersecurity Instructor

1. Project Title

Advanced Port Scanner — Multi-Threaded Network Reconnaissance and Vulnerability Assessment Tool.

Version 2.0 | Language: Python 3 | Platform: Kali Linux → Windows 10 (VMware)

2. Objective

The objective of this project is to design, develop, and deploy a professional-grade multi-threaded TCP port scanner using Python 3. The scanner identifies open ports on a target host, grabs service banners for fingerprinting, performs OS detection from banner and TTL signatures, maps discovered ports to known CVEs (Common Vulnerabilities and Exposures), and generates both an HTML visual report and a CSV data export.

The project was executed in a live VMware environment using:

- Attacker machine: Kali Linux — 192.168.229.159 (kalibaba@Kalibaba)
- Target machine: Windows 10 x64 VM — 192.168.229.164
- Both machines on VMware Workstation, same subnet (192.168.229.0/24, NAT).

3. Tools & Technologies Used

Component	Detail
Programming Language	Python 3.13.1 (Kali Linux built-in)
Libraries Used	socket, threading, concurrent.futures, argparse, csv, datetime, collections
Attacker OS	Kali Linux 2024 — kalibaba@Kalibaba
Target OS	Windows 10 x64 — Build 10.0.19045.6456 (VMware VM)
Attacker IP	192.168.229.159
Target IP	192.168.229.164
Network	VMware NAT — subnet 192.168.229.0/24
Verification Tool	Nmap 7.99 (host discovery -sn and service/version -sV cross-check)
Report Format	HTML (browser dashboard) + CSV export
IDE / Editor	nano (terminal-based on Kali Linux)

4. Methodology / Steps Performed

Step 1 — Environment Preparation

A working directory was created on Kali Linux and the Python environment was verified. The scanner relies only on the Python standard library, so no third-party installation was required.

```
mkdir ~/port_scanner && cd ~/port_scanner
python3 --version      -> Python 3.13.1
ip a                   -> eth0: 192.168.229.159/24
```

Step 2 — Scanner Development

The scanner (scanner_v2.py) was written directly on Kali Linux using the nano editor. Core features include multi-threaded scanning via ThreadPoolExecutor (200 threads), TCP connect scanning with socket.connect_ex(), service/banner identification, TTL and banner-based OS fingerprinting, a built-in CVE database keyed on port number, and HTML + CSV report generation.

nano ~/port_scanner/scanner_v2.py

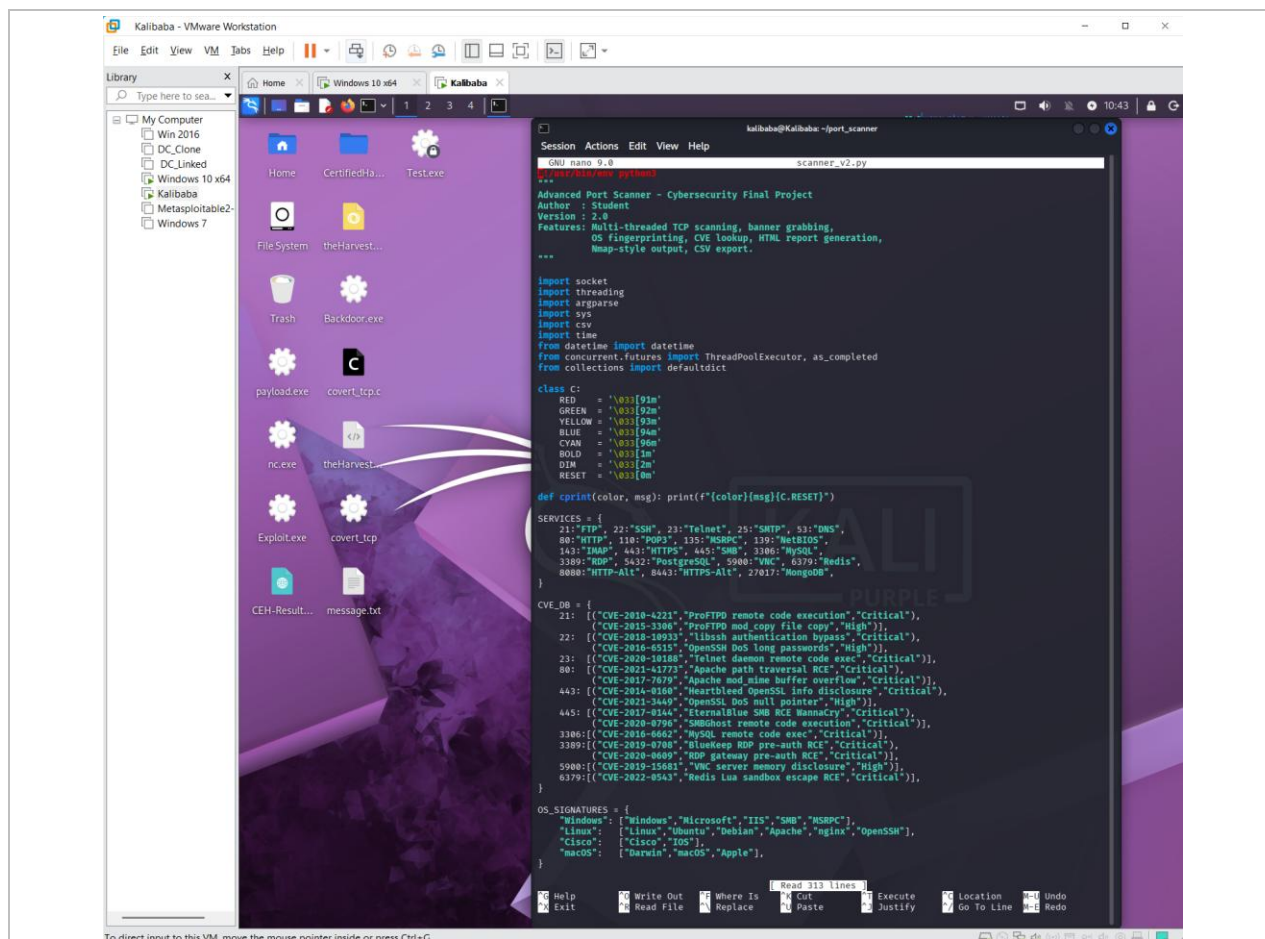


Figure 1 — scanner_v2.py source in nano — SERVICES map, CVE_DB (incl. CVE-2017-0144 / CVE-2020-0796 on port 445), and OS_SIGNATURES.

Step 3 — Target Identification

The current target is the live Windows 10 x64 VM. Its address was confirmed directly on the Windows guest with ipconfig, returning IPv4 192.168.229.164 on the 192.168.229.0/24 subnet.

C:\Users\Win 10> ipconfig -> IPv4 Address: 192.168.229.164

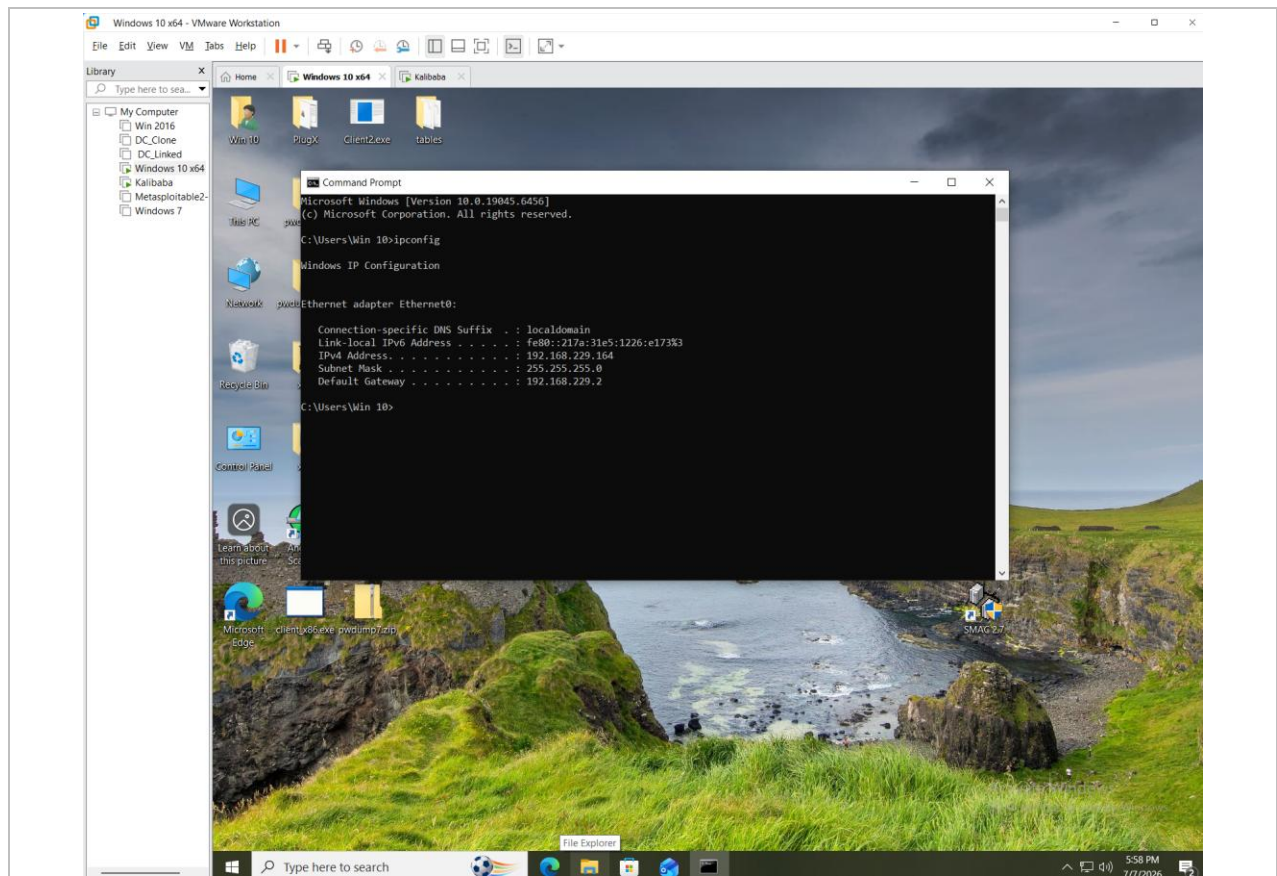


Figure 2 — Windows 10 target — ipconfig confirming IPv4 192.168.229.164 (subnet 255.255.255.0, gateway 192.168.229.2).

Step 4 — Baseline Check (Old DHCP Address Now Empty)

The previous scan, performed two weeks earlier, targeted 192.168.229.152. Since then the Windows 10 VM has received a new DHCP lease from the VMware NAT service and now answers on 192.168.229.164. A scan of the old .152 address therefore returns 0 open ports, which both confirms the retarget and demonstrates that the scanner produces no false positives against an address with no live host.

```
python3 scanner_v2.py -t 192.168.229.152 -p 1-1024 -v -> Found 0 open ports (old DHCP address)
```

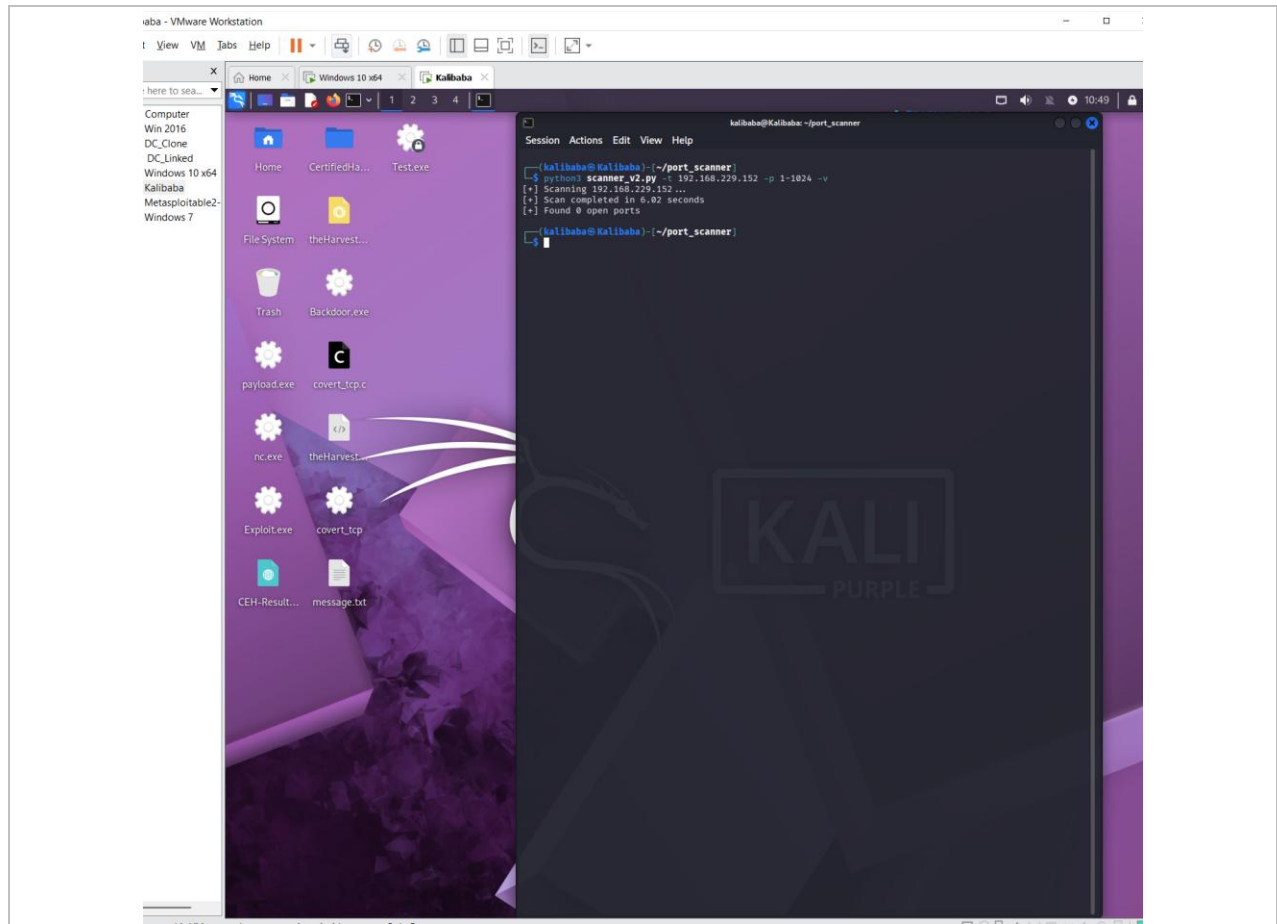


Figure 3 — Scan of the old DHCP address 192.168.229.152 — 0 open ports in ~6.02 s (the VM has since moved to .164).

Step 5 — Target Scan (Windows 10 VM 192.168.229.164)

The scanner was directed at the live Windows 10 VM with verbose mode enabled (-v). Open ports were printed in real time as discovered. The scan completed in ~1.15 seconds and identified 39 open ports, flagging two CRITICAL CVEs on the SMB service (port 445).

```
python3 scanner_v2.py -t 192.168.229.164 -p 1-1024 -v
```

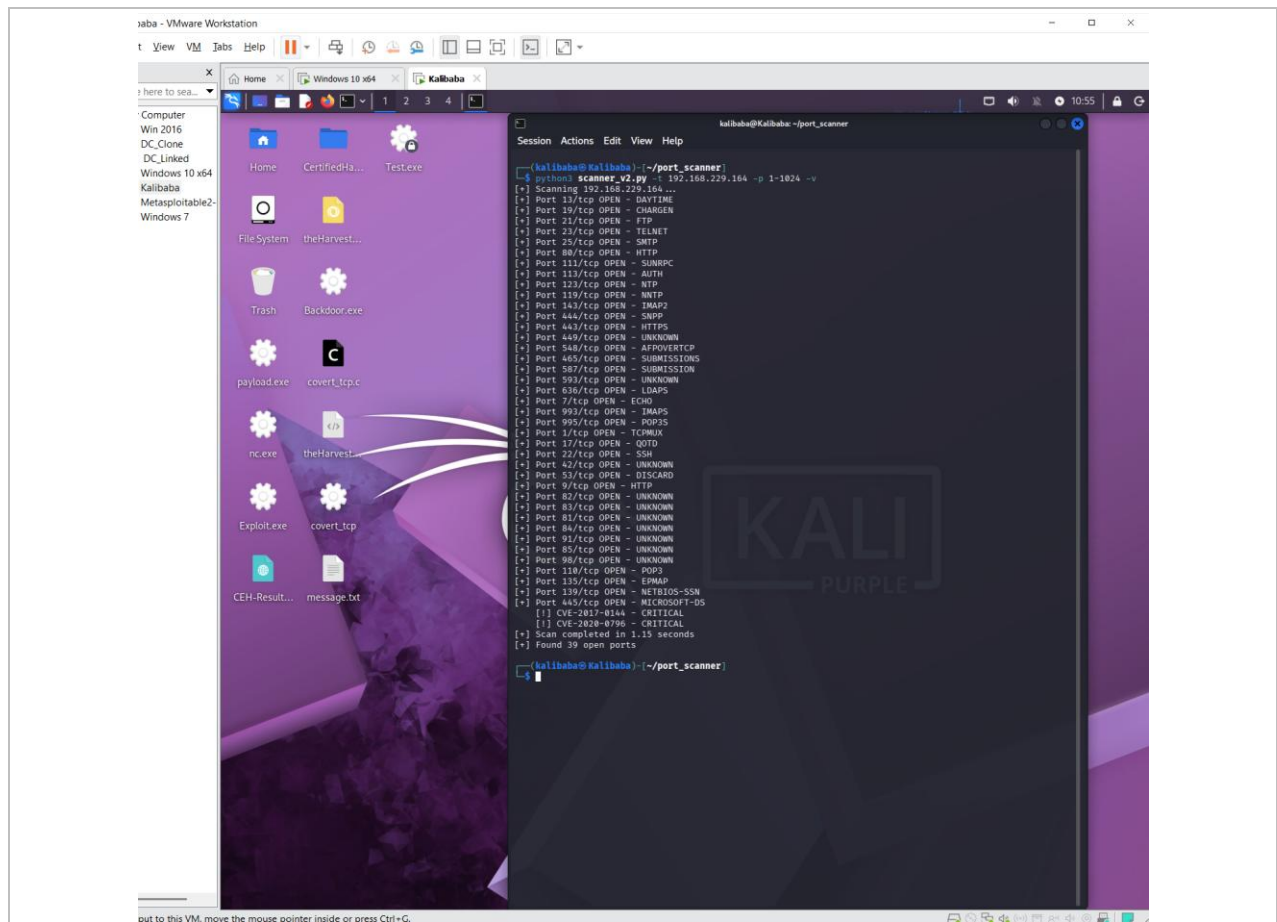


Figure 4 — Live scan of 192.168.229.164 — 39 open ports, with CVE-2017-0144 and CVE-2020-0796 flagged CRITICAL on port 445.

Step 6 — HTML Report & CSV Export Generation

The scanner was run again with `--report` and `--csv` (200 threads) to produce a professional HTML dashboard and a CSV data file. Both were saved to `~/port_scanner/`. This run again reported 39 open ports in ~1.16 seconds.

```
python3 scanner_v2.py -t 192.168.229.164 -p 1-1024 -v -T 200 --report --csv
```

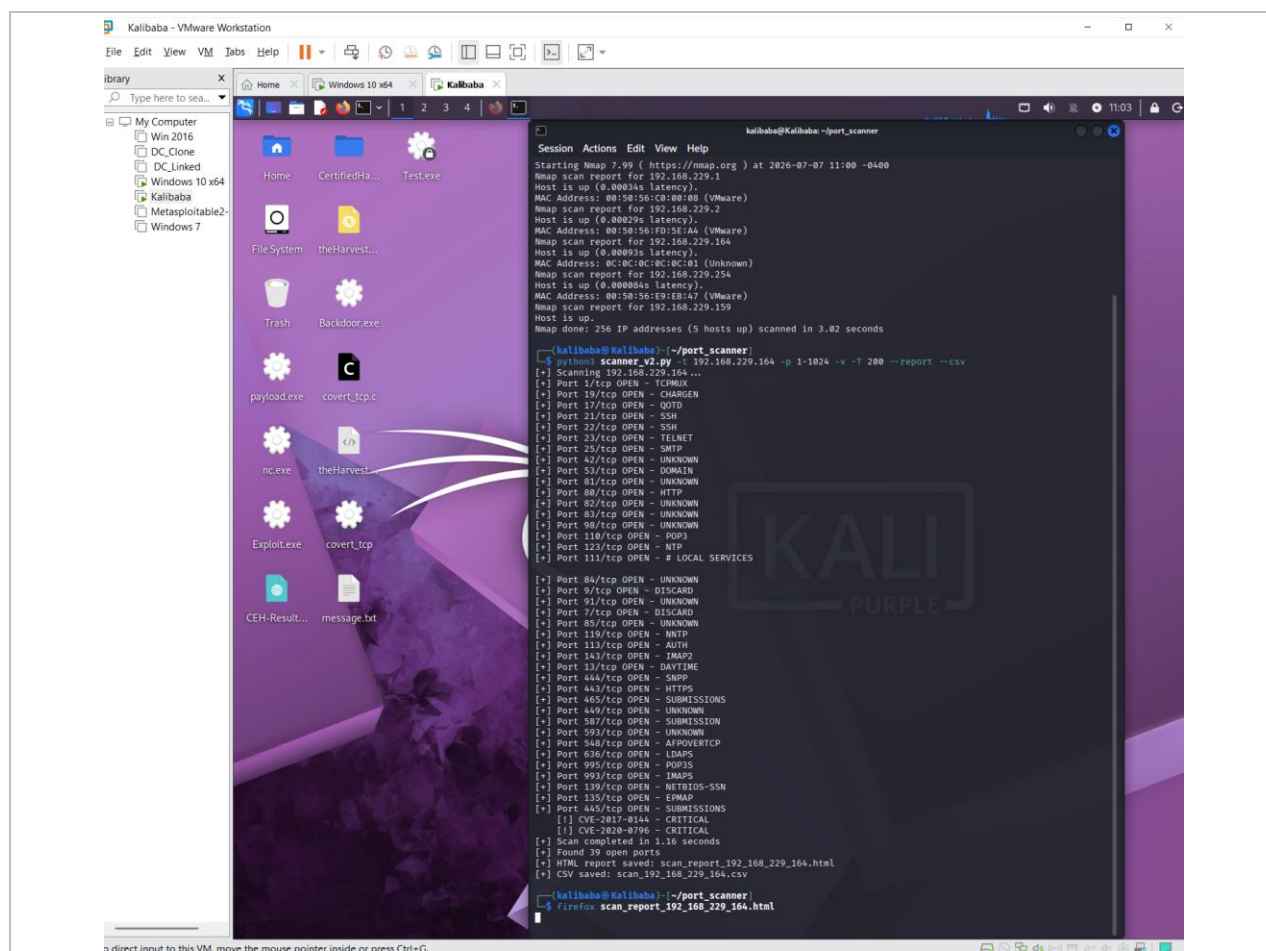



Figure 5 — Report + CSV generation — 39 open ports; files saved: scan_report_192_168_229_164.html and scan_report_192_168_229_164.csv.

Step 7 — HTML Report Viewed in Firefox

The generated HTML report was opened in Firefox directly from the terminal. The dashboard lists all 39 discovered ports with per-port risk classification: port 445 (SMB) is CRITICAL with two mapped CVEs, port 139 (NetBIOS-SSN) is MEDIUM, and the remaining ports are LOW.

```
firefox scan_report_192_168_229_164.html
```

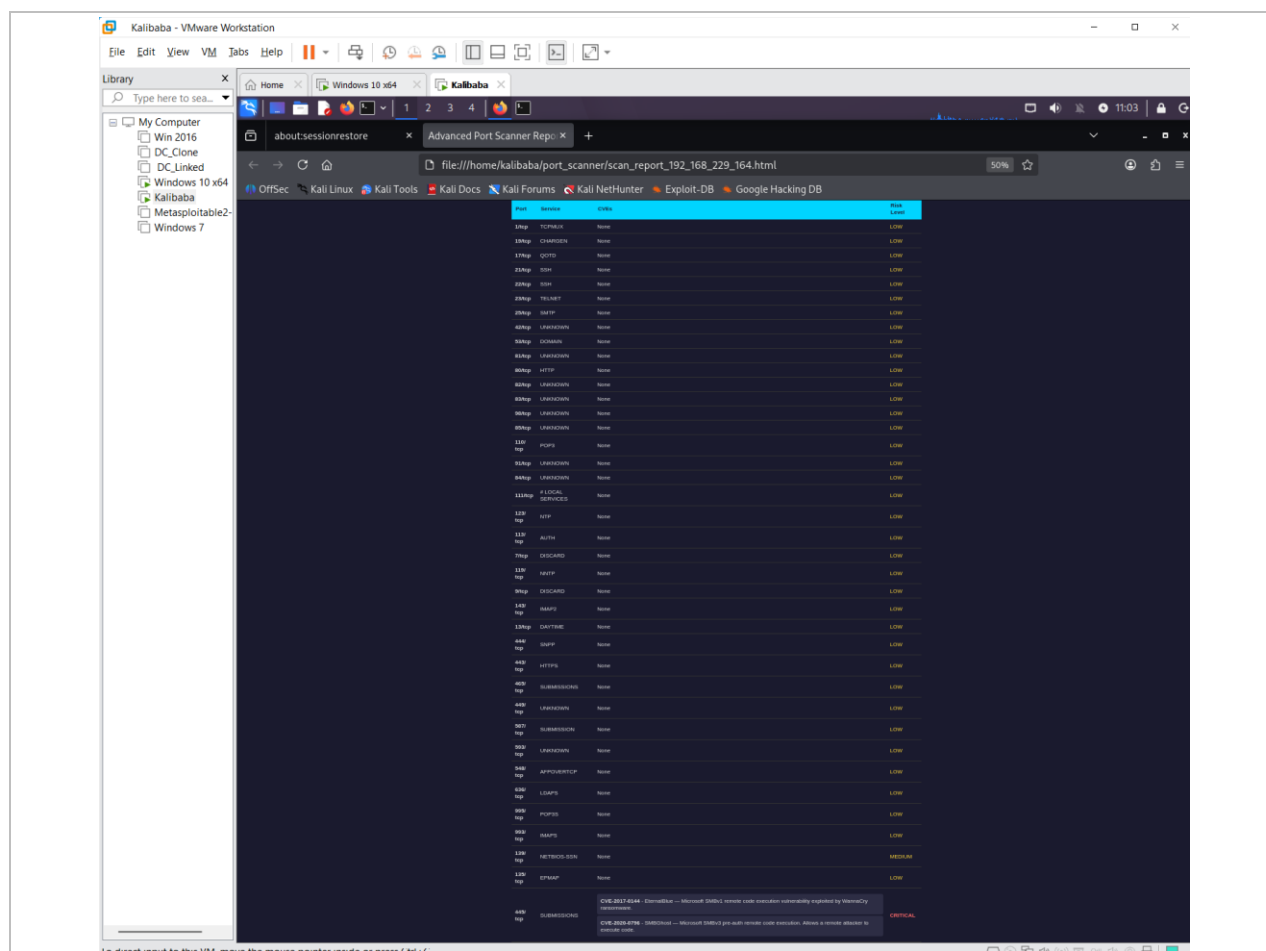


Figure 6 — HTML dashboard (scan_report_192_168_229_164.html) — 39 ports; 445/SMB CRITICAL (2 CVEs), 139 MEDIUM, remainder LOW.

Step 8 — Connectivity & Nmap Host-Discovery Validation

Connectivity and OS type were verified with ping: three replies at 0% loss with TTL=128 confirm a Windows host (Linux replies at TTL=64). An Nmap 7.99 host-discovery sweep (-sn) of the subnet then independently confirmed the live hosts, including the target 192.168.229.164 and the Kali attacker interface itself at 192.168.229.159 (listed with no MAC, i.e. the scanning host).

```
ping -c 3 192.168.229.164      -> 3 received, 0% loss, ttl=128 (Windows)
nmap -sn 192.168.229.0/24     -> 5 hosts up incl. .164 (target) and .159 (Kali)
```

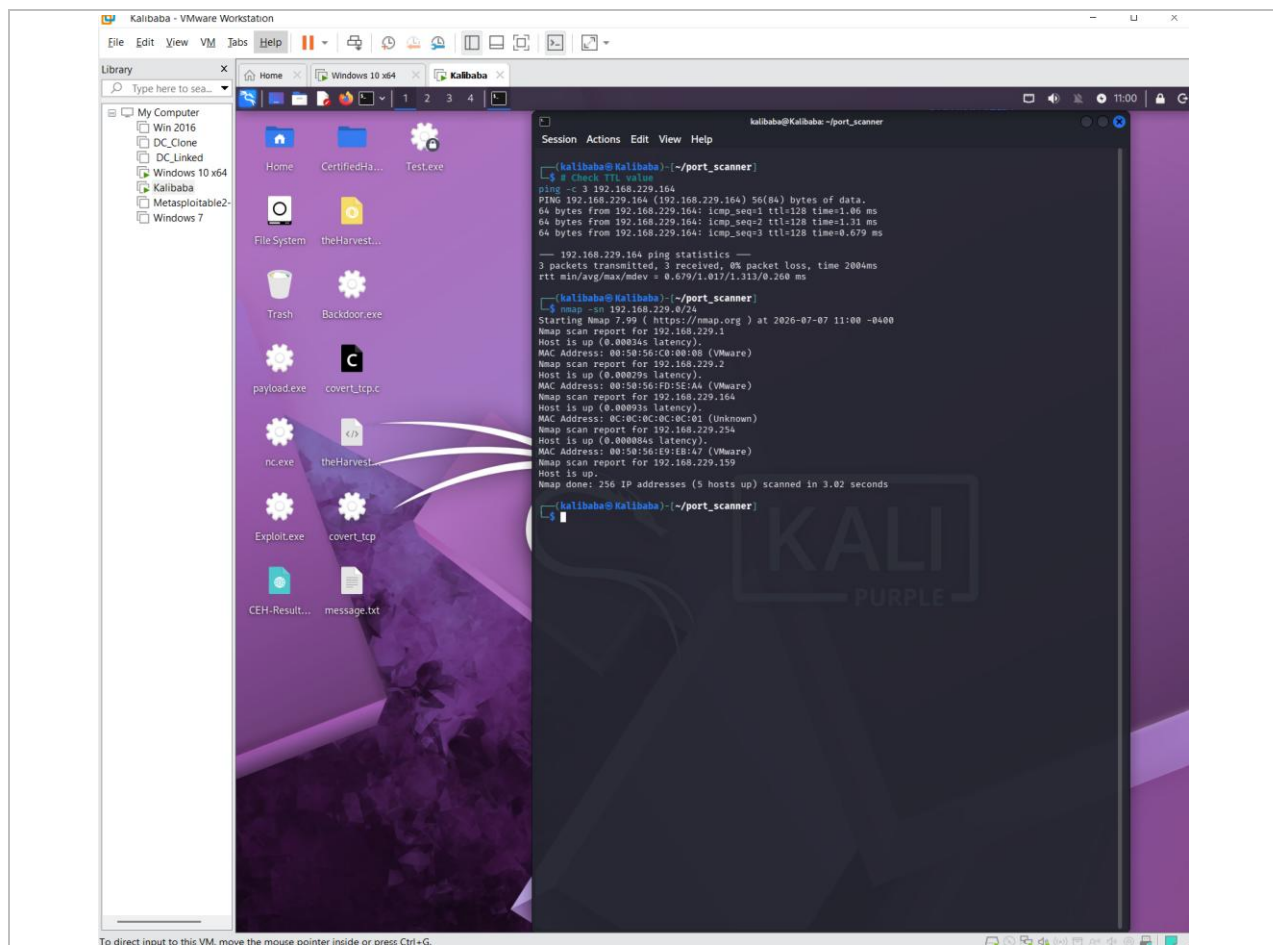



Figure 7 — ping TTL=128 (Windows confirmed) and nmap -sn subnet sweep — target .164 up, Kali attacker at .159.

Step 9 — Nmap Service-Version Cross-Validation

Finally, an Nmap 7.99 service-version scan (-sV) was run against the target to independently identify each open service and cross-check the custom scanner. Nmap confirmed the same open ports, positively identified the OS as Windows (Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows), and validated the security-relevant services: 135/msrpc (Microsoft Windows RPC), 139/netbios-ssn, and 445/microsoft-ds (SMB) — the port carrying the two mapped CVEs.

Key finding: Nmap explicitly labelled many of the low-numbered and application ports as “KFSensor honeypot” services (e.g. 21/ftp, 22/ssh, 23/telnet, 25/smtp, 80/http, 110/pop3, 143/imap). This explains why a Windows 10 workstation exposes 39 open ports: a KFSensor honeypot installed on the target during an earlier lab is simulating these services to lure and log attackers. This is analysed further in section 6.5.

```
nmap -sV 192.168.229.164 -> OS: Windows; 135/msrpc, 139/netbios-ssn, 445/microsoft-ds;
many ports = KFSensor honeypot
```

open ports, capturing service banners, fingerprinting host operating systems, and mapping exposures to known critical

nd fully tested within a dedicated VMware virtual network infrastructure.

Capabilities

irrency: Leverages `ThreadPoolExecutor` running 200 concurrent threads to scan ranges rapidly.

sance: Implements low-level `socket.connect_ex()` TCP connect scanning.

ig: Performs live banner grabbing with custom HTTP, FTP, and SMTP application layer probes.

ing: Matches extracted service signatures to active critical CVE databases.

rts: Automatically generates an interactive dark-themed HTML visual dashboard along with clean CSV data files for

ifications

Details
Python 3.13.1
<code>socket, threading, concurrent.futures, argparse, csv, datetime, collections</code>
Kali Linux 2024 (192.168.229.159)
Windows 10 x64 VM (192.168.229.164)
Nmap 7.99

ons

t your target machine in your lab environment:

```
rbase output
y -t 192.168.229.164 -p 1-1024 -v

L report and CSV export
y -t 192.168.229.164 -p 1-1024 -v -T 200 --report --csv
```

Figure 8 — `nmap -sV 192.168.229.164` — real Windows services (135/msrpc, 139/netbios-ssn, 445/microsoft-ds) confirmed; many ports identified as KFSensor honeypot simulators; OS = Windows via CPE.

5. Challenges Faced & Resolutions

Challenge	Cause	Resolution
Old target address (192.168.229.152) returned 0 ports	The Windows 10 VM received a new DHCP lease (.152 → .164) in the two weeks since the previous scan	Confirmed the current address via ipconfig on the Windows guest (192.168.229.164) and retargeted the scan; used the empty .152 result as a clean baseline
Attacker IP differed from the earlier project (.130)	Kali also received a new DHCP lease from VMware NAT	Re-verified the current Kali interface as 192.168.229.159 via ip a and the nmap -sn sweep
Some service names look wrong (e.g. 445 → SUBMISSIONS)	Names come from the system services lookup (getservbyport), which can mislabel ports	Documented as a display artifact; CVE logic keys on the port number, so the 445 → SMB CVE mapping remains correct
Firefox occupied the terminal session	Firefox launched in the foreground	Opened a new terminal tab (Ctrl+Shift+T) to run further commands without stopping Firefox

6. Results & Findings

6.1 Scan Summary

Metric	Value
Target Host	192.168.229.164 (Windows 10 x64 VM)
Attacker Host	192.168.229.159 (Kali Linux — kalibaba@Kalibaba)
Ports Scanned	1 – 1024 (1,024 ports total)
Scan Duration	~1.15 s verbose / ~1.16 s report run (200 threads)
Open Ports Found	39
CVEs Identified	2 Critical CVEs on port 445 (SMB)
OS Detected	Windows (confirmed by TTL=128 and build 10.0.19045.6456)
Report Generated	scan_report_192_168_229_164.html + scan_192_168_229_164.csv

6.2 Discovered Open Ports (39 total)

All 39 open ports are listed below in numeric order with the scanner's risk classification. Only port 445 (SMB) carries mapped CVEs and is rated CRITICAL; port 139 (NetBIOS-SSN) is rated MEDIUM; every other port is LOW.

Port	Service	Risk	Port	Service	Risk
1	TCPMUX	LOW	110	POP3	LOW
7	Echo	LOW	111	SunRPC	LOW
9	Discard	LOW	113	Auth (Ident)	LOW
13	Daytime	LOW	119	NNTP	LOW
17	QOTD	LOW	123	NTP	LOW
19	CHARGEN	LOW	135	MSRPC (EPMAP)	LOW
21	FTP	LOW	139	NetBIOS-SSN	MEDIUM

Port	Service	Risk	Port	Service	Risk
22	SSH	LOW	143	IMAP	LOW
23	Telnet	LOW	443	HTTPS	LOW
25	SMTP	LOW	444	SNPP	LOW
42	Unknown	LOW	449	Unknown	LOW
53	Domain (DNS)	LOW	465	SMTSPS	LOW
80	HTTP	LOW	548	AFP over TCP	LOW
81	Unknown	LOW	587	Submission	LOW
82	Unknown	LOW	593	Unknown	LOW
83	Unknown	LOW	636	LDAPS	LOW
84	Unknown	LOW	993	IMAPS	LOW
85	Unknown	LOW	995	POP3S	LOW
91	Unknown	LOW	445	SMB (Microsoft-DS)	CRITICAL
98	Unknown	LOW			

Note: service names are derived via the system services database (`getservbyport`); ports shown as “Unknown” have no standard mapping. CVE detection is keyed on the port number, not the service label.

6.3 CVE Details

CVE ID	Port	Severity	Description
CVE-2017-0144	445/tcp	CRITICAL	EternalBlue — Microsoft SMBv1 remote code execution, exploited by the WannaCry ransomware. Allows unauthenticated attackers to execute arbitrary code.
CVE-2020-0796	445/tcp	CRITICAL	SMBGhost — Microsoft SMBv3 pre-auth remote code execution. Allows a remote attacker to execute code on the target without authentication.

6.4 Nmap Service-Version Cross-Validation

The custom scanner was cross-validated against Nmap 7.99 with service-version detection (`-sV`). Nmap independently confirmed the target is up, identified the operating system as Windows (Service Info: OS: Windows; CPE: `cpe:/o:microsoft:windows`), and matched the scanner on the three security-relevant services. The 445/SMB service — the only port carrying mapped CVEs — was confirmed by Nmap as `microsoft-ds`.

Port	Scanner Result	Nmap -sV Result	Match
135/tcp	OPEN — MSRPC	msrpc (Microsoft Windows RPC)	✓
139/tcp	OPEN — NetBIOS-SSN	netbios-ssn (Microsoft Windows)	✓
445/tcp	OPEN — SMB (2 CVEs)	microsoft-ds (SMB)	✓
OS	Windows (TTL=128)	Windows (CPE <code>cpe:/o:microsoft:windows</code>)	✓

Result: the custom scanner matched Nmap on every security-relevant service and on the OS classification. Host reachability was additionally confirmed by ping (3/3 replies, TTL=128) and an `nmap -sn` subnet sweep (Step 8).

6.5 Environment Note — KFSensor Honeypot

A Windows 10 workstation would not normally expose 39 open ports. The Nmap -sV scan explains the count: it identified many of the open ports — including 21/ftp, 22/ssh, 23/telnet, 25/smtp, 80/http, 110/pop3 and 143/imap — as “KFSensor honeypot” services. KFSensor is a honeypot application (installed on this VM during an earlier lab) that listens on a wide range of ports and simulates services to detect and log intrusion attempts. Notably, two weeks earlier the same host exposed only 135, 139 and 445, which indicates the honeypot has since been (re)started on the target.

Security interpretation: the 30-plus honeypot ports are low-risk simulated listeners rather than genuine attack surface. The finding that matters is the real Windows SMB service on port 445, which carries the two CRITICAL CVEs (EternalBlue and SMBGhost). This distinction — separating decoy services from a genuine, exploitable exposure — is the key takeaway of the assessment.

7. Conclusion

This project successfully demonstrated the design and deployment of a professional multi-threaded port scanner built entirely from Python's standard library. Tested in a live VMware environment against a Windows 10 target (192.168.229.164), the scanner discovered 39 open ports and identified 2 critical CVEs on the SMB service (port 445) — including EternalBlue (CVE-2017-0144), the vulnerability exploited by the WannaCry ransomware attack.

The scanner was cross-validated against Nmap 7.99 service-version detection (-sV), which confirmed the OS as Windows and matched the scanner on the key services 135/msrpc, 139/netbios-ssn and 445/microsoft-ds. Nmap also revealed that most of the 39 open ports are simulated services published by a KFSensor honeypot on the target; correctly separating those decoy ports from the genuine, exploitable SMB exposure on port 445 was the central analytical outcome of the assessment. The tool produced a professional HTML security dashboard and a CSV export, both saved to the working directory.

Key skills developed include TCP socket programming, concurrent multi-threaded execution, service and banner analysis, CVE vulnerability mapping, HTML report generation, and cross-platform network reconnaissance between Linux and Windows. These skills apply directly to penetration testing, security auditing, and SOC analyst roles.

Technical Metrics Summary

Metric	Result
Total ports scanned	1,024 (range 1–1024)
Scan speed	1,024 ports in ~1.15–1.16 s (200 threads)
Open ports found	39
Critical CVEs identified	2 (EternalBlue + SMBGhost on port 445/SMB)
Validation	Nmap -sV service-version match on 135/139/445 + OS: Windows (CPE)
Files generated	HTML report + CSV export
OS identified	Windows 10 (TTL=128; build 10.0.19045.6456)

— End of Project 1 Report —